Help Knowledge – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** TaskFlow Assistant (TFA) · **Child Module:** 09.3 Help Knowledge **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

`HelpKnowledgeBase` (singleton) holds 19 embedded markdown topics with front-matter (title, roles, others, keywords, link). It provides keyword-scored search and role-aware redaction. It has no endpoint and no database: it is reached only through the `search_help` model tool (`SearchHelpTool`, 09.2) inside `POST /api/v1/assistant/conversation/messages` (09.1). Answers are rendered in the normal chat feed by the frontend.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Service – Help	`TaskFlow.Api/Services/Assistant/HelpKnowledge/`	`IHelpKnowledgeBase.cs`, `HelpKnowledgeBase.cs`, `HelpTopicDto.cs`
Help content	`TaskFlow.Api/Services/Assistant/Knowledge/`	19 `*.md` topics (`about-tfa.md`, `application-settings.md`, `board.md`, `change-password.md`, `change-role.md`, `change-task-status.md`, `compose-mail.md`, `create-task.md`, `department-management.md`, `edit-profile.md`, `employee-management.md`, `mail-folders.md`, `notifications.md`, `reassign-task.md`, `reminder-settings.md`, `reports-and-workload.md`, `task-comments-and-activity.md`, `task-list-and-filters.md`, `tfa-settings.md`)
Tool (caller)	`TaskFlow.Api/Services/Assistant/Tools/`	`AssistantReadTools.cs` (`SearchHelpTool`, documented in 09.2)
Project file	`TaskFlow.Api/`	`TaskFlow.Api.csproj` (`EmbeddedResource` for the topics)
DI	`TaskFlow.Api/Config/`	`ServiceRegistration.cs` (`AddSingleton` `IHelpKnowledgeBase` → `HelpKnowledgeBase`)

4. Frontend File Structure

Type	Path	Purpose
Child components	`src/app/modules/assistant/components/tfa-message.component.ts/.html`	Renders the answer text and screen link buttons
Child components	`src/app/modules/assistant/components/tfa-composer.component.ts/.html`	Starter suggestion *How do I change a task status?*

No help-specific frontend files.

5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
–	–	(no endpoint)	Used only through the `search_help` tool inside `POST /api/v1/assistant/conversation/messages`	`perm:assistant.chat` (on the send endpoint)	–	`HelpTopicDto` (internal)

5.2 Request / Response Samples

```
// Topic file format: Services/Assistant/Knowledge/create-task.md
---
title: Create a task
roles: Manager, Administrator
others: Tasks are created by Managers and Administrators. If you need a new task, ask your
manager to create and assign it to you.
keywords: create task, new task, add task, assign task, create a task, make a task
link: /tasks/new
---
(markdown body with the steps)
```

```
// search_help result passed to the model (Employee asking "how do I create a task")
{ "found": true, "topics": [ { "title": "Create a task",
  "content": "Tasks are created by Managers and Administrators. If you need a new task, ask
your manager to create and assign it to you.",
  "notAvailableForThisUsersRole": true } ] }
```

5.3 Service Layer Logic

Method: `HelpKnowledgeBase.Search(string query, string role, int maxResults = 3)`

1. Topics are loaded once (singleton) from embedded manifest resources whose name contains `.Assistant.Knowledge.` and ends `.md`; `Topic.Parse` reads the `---` front-matter (`title`, `roles` default `all`, `others`, `keywords`, `link`).
2. Query is lower-cased and tokenised (separators, length ≥ 3, stop words removed).

3. Score per topic: multi-word keyword contained in query +6; single keyword whole-word match (plural tolerant) +4; token in keyword words +3; token in title words +2.
4. Keep score ≥ 3 (`MinimumScore`), order by score desc then id, take 3.
5. `ToDto`: if `roles` contains `all` or the caller's role name → full body + link; otherwise `others` note (or "{Title} is not available to your role.") with no link and `IsRestricted = true`.

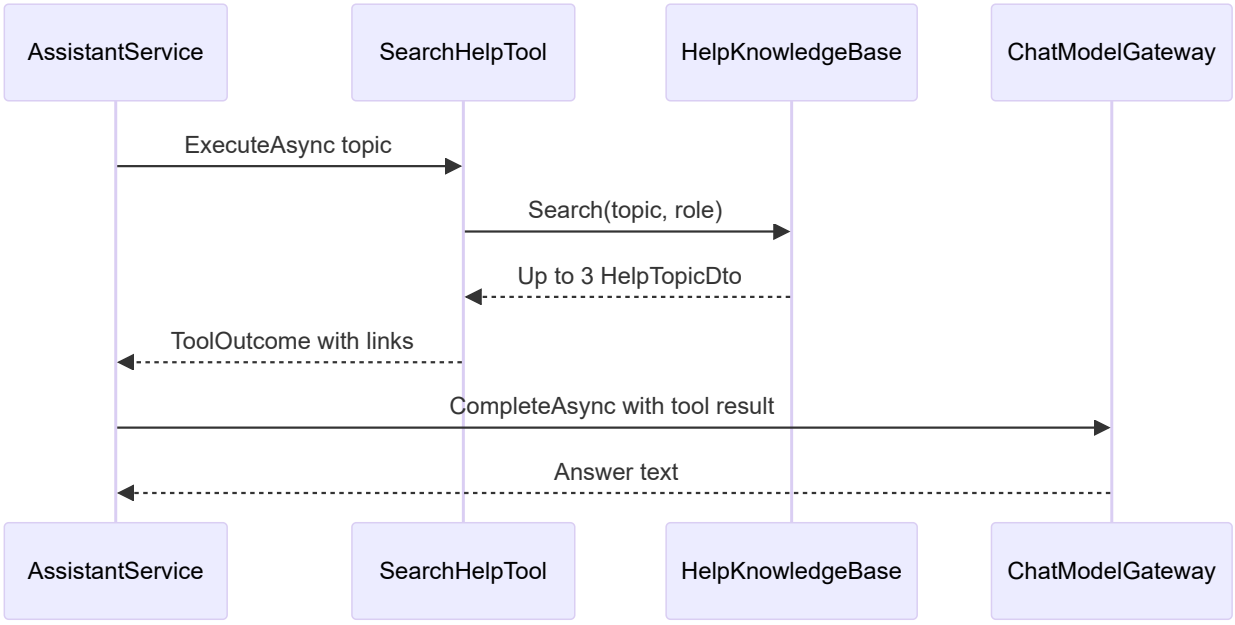
> ✅ Confirmed: `TaskFlow.Api.csproj` declares `<EmbeddedResource Include="Services\Assistant\Knowledge\*.md" />`, so `LoadEmbedded` finds the help files.

Topic `roles:` values in the shipped files: `Administrator` – `application-settings`, `change-role`, `department-management`, `employee-management`, `tfa-settings`; `Manager, Administrator` – `create-task`, `reassign-task`, `reminder-settings`, `reports-and-workload`; `all` – the other 10.

5.4 Repository Layer

Method	Description	Query Notes
–	No database access	In-memory list of embedded topics

5.5 Sequence Diagram



---

6. Database Design

Not applicable – Help Knowledge uses no database tables; topics are embedded resources held in memory.

7. Security & Authorization

- **Authorization:** reached only through `POST /api/v1/assistant/conversation/messages` (`perm:assistant.chat`); the `search_help` tool itself has no extra policy.
- **Data scope:** role-aware redaction – a topic whose `roles` do not include `all` or the caller's role claim name returns only the `others` note, without steps or link.
- **Frontend:** visible as part of the TFA Panel (holders of `assistant.chat`).

Permission Key	Endpoint(s)	UI Element Controlled
`assistant.chat`	`POST /api/v1/assistant/conversation/messages` (tool `search_help`)	How-to answers and link buttons in the TFA Panel

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
No topic scores ≥ 3	`HelpKnowledgeBase.Search`	200 (tool result, no topics)	Model tells the user it does not have that information
Topic not for caller's role	`HelpKnowledgeBase.ToDto`	200 (tool result, `IsRestricted`)	`others` note, or "{Title} is not available to your role."

9. Integrations / Side Effects

- **Notifications / Email / Activity Log:** none.
- **Other modules:** topic content describes screens of other modules and links to their routes (for example `/tasks/new`, `/profile`).

10. Configuration

Not applicable – no configuration keys. Code constants: `HelpKnowledgeBase.MinimumScore` 3; `Search maxResults` default 3.

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Assistant/HelpKnowledge/HelpKnowledgeBaseTests.cs`	Common questions (status change, mail, profile, password, notifications, "what can you do") find their topic, and the status-change topic has the steps and a `/tasks` link; Employees asking about departments, role changes or Manager-only topics get a restricted topic that says who can do it (for departments: no steps and no link), while Administrators and Managers get the steps; unrelated or empty questions return nothing; password help says not to type a password into the chat and links to `/profile`; the result count is limited
Unit	`tests/TaskFlow.UnitTests/Assistant/Tools/AssistantReadToolsTests.cs`	`search_help` part (shared with 09.2): answers from the knowledge base with a `/tasks` link; for a role that cannot do it returns only the restriction note (`notAvailableForThisUsersRole`); an unknown topic returns `found: false`

12. Known Limitations / Gaps found

- Help search is keyword scoring over 19 static topics; no semantic search.

Gaps found

- **Help topics are role-name based, not permission based:** `roles:` front-matter (Administrator; Manager, Administrator) is compared with the role claim name, so a custom role that holds e.g. `tasks.create` still gets the "not available to your role" note; and an Employee granted extra permissions likewise.
- **Help topic link mismatch:** `department-management.md` is titled "Project management" (project keywords) but links to `/departments`; a `/projects` route exists.

© 2026 Jojo George. All rights reserved. Unauthorized copying or distribution of this document is prohibited.